CS341 · COMPUTER GRAPHICS

Midterm Review

The whole midterm in one file

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What's inside

01 The exam distribution

02 Introduction

03 Chapter 1: displays

04 Short answer bank

05 Chapter 2: vectors

06 DDA: (2,2) to (4,10)

01 The exam distribution

Part	Marks	From
True / false	**5**	All chapters
Multiple choice	**5**	All chapters
Short answers: one list, one reason, two comparisons	**4**	Mostly Chapter 1
DDA problem: calculations, the full table and the drawing	**5**	Output Primitives
Aspect ratio problem	**1**	Chapter 1

- No code and no CRT figure.
- The scope ends at slide 44 of Output Primitives (the end of DDA).
- Short questions come straight from the slides, e.g. "Why do we use aspect ratio?"

02 Introduction

Question	Answer
Define CG	**Creation, storage, and manipulation of models and images**
Main tasks	Modeling · Rendering · Animation
NOT a main task	Data encryption
Applications	CAD · Medical · Entertainment → **All of the above**
"Used only for entertainment"	**False**
Goal	Synthetic images indistinguishable from reality

03 Chapter 1: displays

Term	Answer
Random scan	The beam draws lines directly · picture in the **display list**
Raster scan	Scans top to bottom, line by line · picture in the **frame buffer**
A CRT emits	**Electrons**
Persistence	Time for the light to decay to **one-tenth** of its intensity
Refresh rate	How many times per second the picture is redrawn
Resolution	Maximum number of points without overlap
Screen clarity	resolution · **dot pitch** · refresh rate
Dot pitch	Space between centers of adjacent pixels; should be **less than 0.31 mm**
Input device	Mouse (not monitor, printer or speaker)
Hard-copy devices	Laser · Ink-jet · Film recorder · Electrostatic · Pen plotter

Aspect ratio

Why? To maintain geometric features: a square stays a square and a circle does not become an ellipse.

$$\text{AR} = \frac{\frac{\text{vertical points}}{\text{height}}}{\frac{\text{horizontal points}}{\text{width}}}$$

$$461: \quad \text{AR} = \frac{\frac{1080}{5}}{\frac{1920}{10}} = \frac{216}{192} = 1.125$$

Absolute vs Relative

Absolute: the actual position, measured from the origin.

Relative: an offset from the last position (current position). (3, 8) + (2, −1) → (5, 7)

04 Short answer bank

Lists

List	Answer
Input devices	Keyboard · Mouse · Trackball · Joystick · Data glove · Digitizer · Scanner · Touch panel · Light pen · Voice system
Hard-copy devices	Laser printer · Ink-jet printer · Film recorder · Electrostatic printer · Pen plotter
Applications of CG	CAD · Medical visualization · Entertainment · Simulation · Education · Training · Scientific visualization · User interfaces
The three tasks	Modeling · Rendering · Animation
Screen clarity depends on	Resolution · Dot pitch · Refresh rate
Flat-panel advantages	Thinner · Less weight · Less power · No radiation

Reasons

Why?	Answer
Why use aspect ratio?	To maintain geometric features: a square stays a square and a circle does not become an ellipse.
Why are raster lines stair-stepped?	Each point is rounded to the nearest pixel, so all lines except horizontal and vertical look stair-stepped.
Why does random scan need little memory?	It stores only the display program (line-drawing commands), not every pixel.
Why did most systems become raster?	Memory became cheap.
Why is LCD used more than plasma?	Plasma has better resolution but is more expensive.
DDA drawback	Rounding is time consuming, and pixels drift away from the true line on long segments.
Why low persistence for animation?	The light fades fast, so moving pictures do not smear; it needs a higher refresh rate.

Comparisons

	First	Second
Random · Raster	Draws lines directly · display list · very high resolution · limited color	Line by line · frame buffer · stair-step lines · easy color
Absolute · Relative	Actual position from the origin	Offset from the current position
Active · Passive	A transistor per pixel · sharper · more expensive	One per row or column · cheaper · less power
Emissive · Nonemissive	Converts electricity into light: plasma, LED	Uses an outside light source: LCD
Point · Vector	A location in space	Magnitude and direction, no location
Low · high persistence	For animation; needs a higher refresh rate	For complex static pictures

05 Chapter 2: vectors

Statement	Answer
A vector has both magnitude and direction.	**True**
Vector from P to Q	**Q – P**
Sum two points results in a valid point.	**False**
The dot product of two perpendicular vectors is zero.	**True**
A unit vector has a length greater than 1.	**False** (= 1)

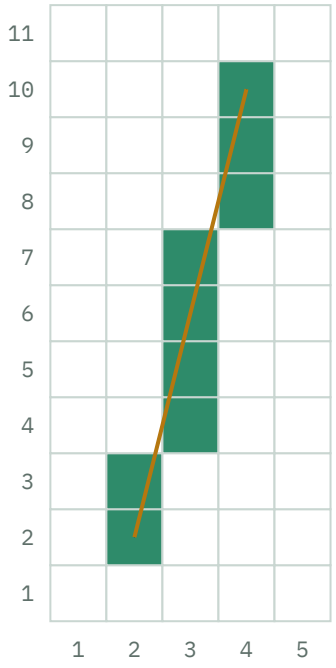
06 DDA: (2,2) to (4,10)

1) dx = 2, dy = 8, steps = the larger = 8

2) $xInc = \frac{2}{8} = 0.25$ $yInc = \frac{8}{8} = 1$

3) round x every step (0.5 rounds up)

k	x	y	round(x)	round(y)
0	2	2	2	2
1	2.25	3	2	3
2	2.5	4	3	4
3	2.75	5	3	5
4	3	6	3	6
5	3.25	7	3	7
6	3.5	8	4	8
7	3.75	9	4	9
8	4	10	4	10



DDA: floating-point increments, then rounding. Rounding is time consuming, and pixels drift away from the true line on long segments.